Carryover representation and inference for biomarker-treatment interaction tests in aggregated N-of-1 trials under covariance moderation: a factorial simulation comparison of nine analysis specifications

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August 19, 2026

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Abstract

Background. Aggregated N-of-1 trials pool repeated within-patient on/off contrasts to validate predictive biomarkers. Under covariance moderation, a multivariate-normal differential-correlation architecture in which the biomarker-treatment interaction lives in a treatment-state-dependent correlation, a companion study found that carryover costs up to roughly 31% of power in relative terms. The analyst must still decide how to represent residual exposure in the linear predictor, and it is not established which representation is most robust to the ordinarily unknown pharmacokinetic decay shape.

Methods. We compare nine analysis-model specifications for the drug-exposure regressor, in two tiers. The Tier 1 comparison, our primary evidence base, covers three specifications across a full factorial grid. Unadjusted is the binary on-drug indicator, ignoring carryover altogether. Lag-adjusted augments that indicator with a lagged just-off-drug term estimating carryover non-parametrically, the classical crossover device. Exposure-weighted is a continuous indicator committing to a parametric decay and half-life, the current package default. We embed these three in an ADEMP-structured factorial simulation[1] crossing two data-generating decay forms (exponential, Weibull) with three trial designs, two sample sizes, and three interaction strengths (216 cells, 500 replicates each), plus sensitivity blocks and a cluster-robust recalibration. All three are fitted to the same simulated datasets, so every contrast is paired, and each performance measure is reported with its Monte Carlo standard error (MCSE). A Tier 2 extension adds six further specifications at a single reference cell: an AIC-selected half-life variant, a paired-difference regression with no repeated-measures model at all, and CR2 cluster-robust variants of four of the above. This second tier is a more targeted comparison, not a full grid.

Results. Three findings stand out. First, the lagged-treatment term buys nothing. Lag-adjusted and Unadjusted are statistically indistinguishable across all 216 Tier 1 cells, differing by a mean of 0.003 in power against an MCSE of 0.018, and agreeing to three decimal places on bias, mean squared error, and coverage. Because the two share an estimand and differ by exactly one nuisance term, this is a clean null. Second, Exposure-weighted attains the highest Tier 1 power only in the two designs with a blinded-discontinuation phase (Hybrid and OL+BDC), where its advantage is about 10 percentage points (0.860 vs 0.766 for Unadjusted at the reference cell). Under the classical crossover (CO) design it is markedly inferior (0.488

vs 0.830). Where it leads, the lead is robust to decay-shape and half-life mis-specification and to autocorrelation strength. The model-based interaction test is mildly conservative for all three (mean null rejection 0.036 to 0.039), an artifact of the working covariance that a CR2 cluster-robust standard error removes without disturbing the ranking. Third, at the Hybrid reference cell the Tier 2 extension finds that combining the AIC-selected half-life with a CR2 standard error outperforms every other specification tested, including the CR2-corrected Exposure-weighted predictor, with Type I error held at nominal despite the post-selection risk the half-life selection step raises in principle. This result held at only that one cell, so we report it as a promising direction rather than a validated recommendation.

Conclusions. Four points warrant emphasis. First, under the covariance-moderation architecture we recommend the exposure-weighted predictor, reported with a cluster-robust standard error, for the Hybrid and OL+BDC designs. Under a crossover design it should not be used, and the unadjusted binary indicator is materially better. Second, the classical lagged-treatment remedy cannot be recommended in any design examined here, since it never improved on ignoring carryover altogether. We trace this to its entering the model as a nuisance main effect, which leaves the attenuation of the interaction untouched. Third, across all conditions anticipated dropout governs power far more than the choice among specifications, and should dominate design effort accordingly. Fourth, the AIC-selected half-life specification with a CR2 correction merits further validation as a candidate default, having outperformed every other specification at the one cell where all nine were compared directly, but it has not yet been tested against the same robustness battery the Tier 1 three have. This manuscript restricts the grid to the covariance-moderation architecture and to the exponential and Weibull decay forms. The mean-moderation architecture and linear decay are evaluated in the archived companion drafts `archive/report.Rmd` and `archive/report-slim.Rmd`.

1 Introduction

In a single N-of-1 trial, one patient is crossed repeatedly between active treatment and a comparator, and the within-patient contrast between on-drug and off-drug periods estimates that patient's individual treatment effect. An aggregated N-of-1 trial pools many such series into a mixed model, trading some within-patient resolution for population-level inference at sample sizes considerably smaller than a parallel-group trial would require[2]. The estimand is the biomarker-treatment interaction, whether a baseline covariate B_i modifies the treatment effect. We examine three trial designs that differ in how on-drug and off-drug occasions are arranged, a classical crossover (CO), an open-label design followed by blinded discontinuation (OL+BDC), and a Hybrid N-of-1 design, each defined in Section 2.2.

The inferential complication specific to this setting is carryover. A drug's pharmacological effect does not switch off abruptly at discontinuation but decays over days or weeks, so nominally off-drug observations still carry residual exposure. An exposure regressor that ignores this is a mis-measured version of what the patient actually received, and the damage is twofold. The interaction contrast is diluted, because occasions still carrying a substantial fraction of the drug effect are coded identically to occasions carrying none, biasing the biomarker-by-exposure coefficient toward zero as any covariate measured with error would. And those same occasions are heterogeneous in a way the model does not represent, so the unexplained variation inflates the standard error of the coefficient that has just been attenuated. The test loses on both counts, and power falls accordingly.

This is a loss of power, not of validity. Under the null, $c_{bm} = 0$, no biomarker-response association exists at any level of residual exposure, so a mis-coded regressor has no interaction to distort and test size is unaffected by exposure coding. The mild conservatism observed in Section 3.6 has a separate origin in the working covariance and applies equally across specifications. Carryover mis-specification is therefore a question of efficiency, and we frame the study in terms of recovered power.

Three answers are available for how to represent residual exposure in the linear predictor, corresponding to three positions an analyst can take toward the unknown decay. The analyst may deny it, coding exposure strictly on or off, may estimate it, adding a nuisance term whose magnitude the data determine, or may assume it, committing to a decay curve and a half-life. A companion study[3], building on Hendrickson[2], shows that under the covariance-moderation architecture, in which the interaction lives in the treatment-state-dependent covariance $\mathrm{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$, carryover erodes power directly and design-dependently. At a half-life of one week, the relative loss is roughly 31% under OL+BDC but only 3 to 5% under crossover and Hybrid, against 1 to 3% under an alternative architecture where the biomarker moderates the mean response directly. We restrict attention here to covariance moderation, where the carryover problem is largest and Hendrickson's original results were obtained. That earlier work fixed the decay shape at exponential and evaluated only one analysis specification, leaving open which representation of residual exposure performs best when the true decay shape is not known precisely, as is ordinarily the case.

We compare three specifications as our primary evidence base. Unadjusted is the binary on-drug indicator, ignoring carryover entirely and serving as the benchmark any remedy must beat. Lag-adjusted augments it with a lagged just-off-drug term that estimates carryover magnitude without assuming a functional form, following the classical crossover-trial device of Jones and Kenward[4]. Exposure-weighted is a continuous indicator committing to a parametric decay form and half-life, the current pmsimstats-ng default. Unadjusted and Lag-adjusted estimate the same coefficient and are fitted to the same simulated datasets, differing by exactly one nuisance term, so their contrast isolates that term's contribution alone. Exposure-weighted differs on a second axis as well, since it changes the regressor carrying the interaction. We cross all three against exponential and Weibull data-generating decay forms and ask whether one can be recommended outright, or whether the choice depends on trial design, following the ADEMP reporting framework of Morris, White, and Crowther[1] throughout. Six further specifications, an AIC-selected half-life variant, a paired-difference regression, and CR2 cluster-robust variants of four base specifications, are added later at a single reference cell (Section 2.5) once the results from these first three raise specific new questions the Tier 1 grid alone cannot answer. Section 2 describes the simulation design in ADEMP order, Section 3 gives the results, and Section 4 takes up the practical implications.

2 Simulation design

The study simulates complete aggregated N-of-1 trials and analyzes each one under every specification. A single replicate proceeds as follows. Patients are allocated across the randomization paths of the chosen trial design (Section 2.2). For each patient we draw a pre-treatment biomarker together with the three latent response components, namely the pharmacological response, the natural-history trend and the placebo-belief component, jointly across all eight measurement occasions from a multivariate normal distribution whose means follow modified Gompertz trajectories and whose covariance carries within-component AR(1) autocorrelation, cross-component correlations, and the treatment-state-dependent biomarker-response correlation that encodes the interaction (Section 2.3). The observed symptom score at each occasion is the patient's baseline severity less the sum of the three components, so that a beneficial effect lowers the score. Carryover enters through a decay function of time since discontinuation, which keeps part of the pharmacological effect present at nominally off-drug occasions. All three analysis specifications (Section 2.4) are then fitted to that same simulated dataset, and we record the interaction coefficient, its standard error and its p-value from each. This is repeated 500 times for every parameter combination in the grid of Section 2.7.

Fitting all specifications to a common dataset is deliberate. It makes every contrast among them paired, so that differences are estimated with substantially less Monte Carlo variance than independent draws would give. Throughout, N denotes the **total** number of patients in a trial, allocated as evenly as possible across the design's randomization paths, so a design with P paths assigns about N/P patients to each. At $N = 70$ this is 35 per path under the two-path designs and 17 or 18 per path under the four-path Hybrid design.

2.1 Aims

The aim is to quantify, under covariance moderation, the cost of mis-specifying the carryover representation in the analysis model, and to determine which of the three candidate strategies performs better across the grid of data-generating decay forms and trial designs. A second aim, addressed in the Tier 2 sensitivity blocks of Section 3.7, is to check whether the resulting ranking survives perturbation of within-subject autocorrelation, dropout rate, half-life accuracy, and interaction size.

2.2 Trial designs

The three designs differ in how on-drug and off-drug occasions are arranged, and so in how much post-discontinuation information each generates. All three schedule eight measurement occasions per patient across randomization paths, each a fixed on-drug/off-drug schedule, and each design carries an expectancy weight (one during open-label periods, one half during blinded periods) scaling the placebo-belief component.

Crossover (CO). A traditional two-period within-patient crossover, blinded throughout, occasions spaced 2.5 weeks apart from 2.5 to 20 weeks. One path is active for the first four occasions then comparator, and the other reverses that order and so never discontinues, contributing no post-discontinuation observations.

Open-label followed by blinded discontinuation (OL+BDC). Open-label titration at 4, 8, 12, and 16 weeks, then weekly follow-up to week 20 with a blinded late switch to placebo. The two paths discontinue after week 17 or 18, so only two or three occasions follow.

Hybrid. The Hendrickson design and this program's reference. Open-label titration at weeks 4 and 8 is followed by blinded discontinuation weekly from week 9 to 12, then a brief two-occasion crossover at weeks 16 and 20. The four paths vary discontinuation timing and crossover-arm order. It is the richest of the three in on-off transitions and the only one allowing more than one discontinuation per patient.

One consequence bears directly on the results. The first post-discontinuation occasion falls one week after switching under OL+BDC and Hybrid, but 2.5 weeks under CO. At the half-lives examined here, roughly half the pharmacological effect therefore persists into the first off-drug occasion under the two non-crossover designs, against less than a fifth under CO, so the designs differ substantially in how much carryover contamination they can accumulate at all (Section 4.2).

2.3 Data-generating mechanism

Under covariance moderation, the biomarker and the biological response are drawn jointly from a multivariate normal distribution with a treatment-state-dependent correlation, $\mathrm{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$, where ϕ is the decay form and t_{sd} the time since drug discontinuation. Mean moderation and the combined dual-channel architecture fall outside this manuscript's scope. The full cross-architecture comparison is in the archived full-scope drafts (`archive/report.Rmd`, `archive/report-slim.Rmd`). The decay function $\phi : \mathbb{R}_+ \to [0, 1]$ maps t_{sd} to a residual-effect multiplier, and we evaluate two forms here, each matched to a common half-life $t_{1/2}$. A third, linear decay, appears in the full-scope manuscripts but not here.

2.3.1 Exponential decay

$$\phi_{\mathrm{exp}}(t_{sd}) = \exp\left(-\lambda t_{sd}\right), \qquad \lambda = \frac{\ln 2}{t_{1/2}}$$

This is the decay form used throughout the companion manuscript and implemented in `functions.R` under `implementations/tidyverse/`. It corresponds to classical first-order elimination kinetics, the default pharmacokinetic assumption, and serves as the reference against which Weibull is compared.

2.3.2 Weibull decay

$$\phi_{\mathrm{weib}}(t_{sd}) = \exp\left(-\left(\lambda_w t_{sd}\right)^k\right), \qquad \lambda_w = \frac{(\ln 2)^{1/k}}{t_{1/2}}$$

A shape parameter k governs whether the elimination tail is heavier than exponential ($k < 1$, slower elimination) or lighter ($k > 1$, faster clearance), recovering the exponential form exactly at $k = 1$. We report $k = 0.5$ (a markedly prolonged low-concentration tail) and

$k = 2.0$ (markedly faster clearance once concentration falls below some threshold), a wider sensitivity range than an earlier version of this grid used ($k = 0.7, 1.5$). $k = 1.0$ is not a third Weibull condition: it is numerically indistinguishable from Exponential and so is not reported separately, giving three decay-shape levels in the grid rather than four.

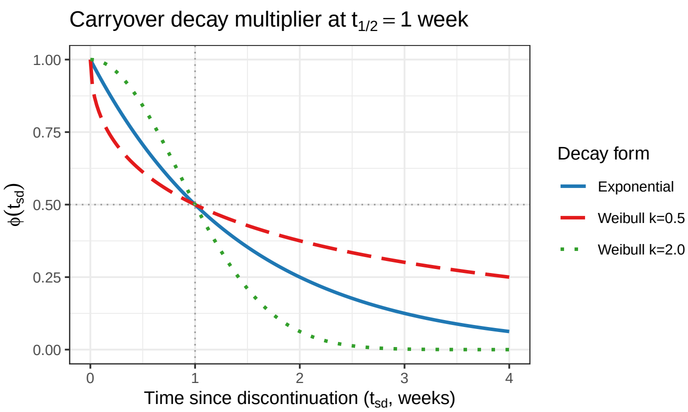

Figure 1: The carryover decay multiplier $\phi(t_{sd})$ at the reference half-life ($t_{1/2} = 1.0$ week) for exponential decay and the two Weibull shapes. Both dotted reference lines mark the defining half-life point ($\phi = 0.5$ at $t_{sd} = t_{1/2}$), which all three curves pass through by construction; the shape parameter governs only how quickly ϕ falls before that point and how slowly it decays after.

Figure 1 makes the shape difference concrete. The $k = 0.5$ curve has a much heavier tail, and ϕ stays well above the exponential curve for every $t_{sd} > t_{1/2}$. In practical terms, the analysis-side predictor picks up substantially more residual carryover during a prolonged washout than the half-life alone would suggest. The $k = 2.0$ curve has a much lighter tail and clears sharply faster once concentration falls below the half-maximum threshold. All three curves pass through the same half-life-defining point ($\phi = 0.5$ at $t_{sd} = 1.0$), so the three forms remain comparable at a fixed nominal half-life even though their shapes diverge substantially on either side of it.

2.4 Estimand and analysis specifications

The interaction estimand is the on-drug biomarker-response correlation c_{bm}, a dimensionless quantity. Bias and empirical SE are reported on a calibrated effect-size scale (the expected difference in on-drug response per SD of the biomarker at $t_{1/2} = 0$, per `docs/19-mean-moderation-implementation-notes.tex`). Power and Type I error are on the native test-statistic scale, with the null estimand $c_{bm} = 0$. The analysis model is `nlme::lme(Sx ~ bm * X + t, random = ~1 | ptID, correlation = corCAR1(form = ~ t | ptID))` for the Tier 1 three, which differ only in how X is constructed and whether a nuisance term is added. Unadjusted and Lag-adjusted both test the coefficient on bm:D_{it}, so every performance measure is comparable between them. Exposure-weighted tests bm:$D_{bc,it}$, a different quantity. Power and Type I error remain comparable across all three, since they are properties of the test, but bias, MSE, and coverage are not comparable across that divide (Section 3.1). Nine specifications are evaluated in total across this manuscript: the Tier 1

Table 1: All nine analysis specifications evaluated in this manuscript. G1-G3 are the Tier 1 headline three (Section 2.4); G4-G5 are Tier 2 additions replacing the retired E4/E5/E6 (Blocks S7-S9, Section 3.7); G6-G9 are CR2 cluster-robust variants of G1, G2, G4, and G3 respectively (Block S10). G10 (G5+CR2) does not apply and is omitted: G5 collapses to one row per patient before fitting, leaving no repeated-measures cluster structure for CR2 to correct.

Code	Stored	Name	Formula	Target coefficient
G1	E1	Unadjusted	Sx bm + t + Db + bm:Db	bm:Db
G2	E3	Lag-adjusted	Sx bm + t + Db + bm:Db + L	bm:Db
G3	E2	Exposure-weighted	Sx bm + t + Dbc + bm:Dbc	bm:Dbc
G4	E7	AIC-selected t1/2	G3 formula, t1/2 by AIC over {0.25,0.5,1.0,2.0}	bm:Dbc
G5	E9	Paired-difference	diff bm (patient-level OLS)	bm slope
G6	E1cr2	Unadjusted + CR2	G1 formula, CR2 SE	bm:Db
G7	E3cr2	Lag-adjusted + CR2	G2 formula, CR2 SE	bm:Db
G8	E2cr2	Exposure-weighted + CR2	G3 formula, CR2 SE	bm:Dbc
G9	E7cr2	AIC-selected t1/2 + CR2	G4 formula, CR2 SE	bm:Dbc

three below, plus six Tier 2 additions introduced in the 'Choice of analysis specifications' section that follows and tested in Blocks S7-S10 (Section 3.7). Internally, and in the code and data files described in the Reproducibility section, each specification is stored under an `E`-prefixed code (`E1`, `E2`, …); this manuscript's prose refers to all nine uniformly by the compact `G1-G9` code defined in `spec-labels.R`, or by name, per Table 1 below.

2.4.1 Unadjusted (G1): binary on-drug indicator

Here $X = D_{it}$, the binary on-drug indicator (one on drug, zero off). Carryover is not represented at all. An off-drug occasion still carrying most of the pharmacological effect is therefore coded identically to one carrying none. This is the default an analyst obtains without considering carryover, and we treat it as the benchmark the other specifications must justify their additional structure against. It consumes no half-life, so it is invariant to the analyst's pharmacokinetic assumptions.

2.4.2 Lag-adjusted (G2): binary indicator plus estimated carryover term

We retain $X = D_{it}$ and augment the model with a lagged-treatment indicator, $L_{it} = D_{i,t-1}(1 - D_{it})$, marking off-drug timepoints that immediately follow an on-drug one. This is the classical crossover-trial device of Jones and Kenward[4]. Carryover magnitude is estimated from the data rather than assumed, so, like Unadjusted, this specification is invariant to the assumed half-life. L_{it} enters only as a nuisance main effect; there is no bm:L_{it} product, so the interaction actually tested remains bm:D_{it}. Lag-adjusted can absorb mean-level carryover, but it cannot repair the interaction's attenuation (Section 1). We think of it as Unadjusted with a carryover nuisance control added, not as a fundamentally different model of the interaction (Section 4.1).

2.4.3 Exposure-weighted (G3): continuous decayed predictor

Here $X = \mathrm{Dbc}_{it}$, a continuous exposure-decayed predictor matched to the analyst's assumed decay form and half-life; this is the current pmsimstats-ng default. At on-drug timepoints

$D_{bc,it} = 1$. Off drug it decays as $D_{bc,it} = (1/2)^{t_{sd,it}/\hat{t}_{1/2}}$, losing half its value every $\hat{t}_{1/2}$ weeks after discontinuation. When the assumed decay form matches the true one, the predictor is well matched. When it does not, it is mis-specified, and we probe the consequence directly in Block S2. The half-life assumption enters only through the predictor values handed to `lme`, not through the model formula, so Exposure-weighted is in effect a different model for each assumed half-life. It is the only one of the three original specifications that consumes a half-life at all, and this is what the S2 sensitivity contrast (Section 3.7) is designed to probe. A limiting relationship connects Exposure-weighted back to Unadjusted. As $\hat{t}_{1/2} \to 0$, $D_{bc,it} \to D_{it}$, so Unadjusted is exactly the boundary member of the Exposure-weighted family at zero assumed half-life. The S2 sweep is therefore a traverse anchored at Unadjusted, not an independent check.

2.4.4 AIC-selected half-life (G4)

This specification uses Exposure-weighted's formula, $X = \mathrm{Dbc}_{it}$, but chooses $\hat{t}_{1/2}$ from the data rather than assuming it. We refit the model at each of four candidate half-lives (0.25, 0.5, 1.0, 2.0 weeks) and keep the fit with the lowest AIC[5]. This mirrors `characterize_carryover()` (`R/carryover_analysis.R`), the package's own real-data half-life-selection routine. Like Exposure-weighted, it targets bm:$D_{bc,it}$. Unlike Exposure-weighted, it requires no half-life from the analyst at all. The cost is a post-selection inference risk[6]: the same data drive the half-life choice and the subsequent test. Block S11 (Section 3.7) checks this risk directly.

2.4.5 Paired-difference (G5)

This is a two-stage, correlation-structure-agnostic alternative, rather than an elaboration of the mixed-model family above. For each patient with both on- and off-drug observations, we compute the patient's own mean on-drug minus mean off-drug S_x, and regress that single per-patient difference on bm across patients (diff $\sim$ bm, plain OLS). There is no repeated-measures model, no carryover representation, and no correlation structure of any kind. We view it as a direct biomarker-interaction extension of the paired t-test that Senn, Julious and Araujo[7] found to outperform carryover-adjusted mixed models for ordinary N-of-1 treatment-effect estimation. Patients contributing only on-drug or only off-drug observations (for instance, CO's never-discontinuing path) cannot form a within-patient difference and are dropped.

2.4.6 CR2 cluster-robust variants (G6-G9)

Section 3.6 (Block S6) shows that the model-based interaction test is mildly conservative, and that a CR2 bias-reduced cluster-robust standard error restores nominal size without disturbing the ranking among G1-G3. We use the estimator of Bell and McCaffrey[8], via `clubSandwich`, paired with Satterthwaite degrees of freedom. That correction, `lme+CR2` in Section 2.7, was applied only to G3 there. Block S10 (Section 3.7) extends it to G1, G2, and G4 as well, replacing only the standard error, reference distribution, and p-value on each specification's existing point estimate: G6 (G1+CR2), G7 (G2+CR2), G9 (G4+CR2). G8 (G3+CR2) is the same specification already introduced in Section 2.7 and Block S6, and we carry it into the nine-way G1-G9 comparison at Block S10 without rerunning it. A tenth combination, G5+CR2, does not apply. G5 already collapses to one row per patient before fitting, so there is no repeated-measures cluster structure left for CR2 to correct.

2.5 Choice of analysis specifications

Three considerations bounded which specifications were tested, rather than surveying the wider space of models used in the N-of-1 and crossover-trial literature.

The Tier 1 three (Unadjusted, Lag-adjusted, Exposure-weighted, above) were chosen because each represents a distinct, established position an analyst can take toward an unmeasured or assumed pharmacokinetic decay, not because they exhaust that space. Unadjusted is the default any analyst obtains without considering carryover at all, and so is the necessary benchmark for every alternative. Lag-adjusted follows Jones and Kenward's[4] classical crossover-trial device, the shape-free remedy the pre-mixed-model crossover literature has long recommended. Exposure-weighted is the pmsimstats-ng package's own established default, so it is the specification any user of the package is actually running today, and the manuscript's practical stake in getting its behavior right is correspondingly higher than for the other two.

This left a wide range of alternative model classes untested by design: Bayesian hierarchical N-of-1 models; GEE and other marginal models, compared systematically rather than as a robustness check on one specification's standard error (Section 2.7); randomization and permutation tests; functional or spline-based flexible decay curves; state-space or Kalman-filter treatments of drug concentration; and two-stage 'series of N-of-1 trials' meta-analytic combination. Each is established in the N-of-1 or crossover-trial literature, and none is evaluated here. This is a scoping decision, not an oversight: the manuscript's aim is to compare carryover *representations* within a single, fixed conditional mixed-model framework, matching Hendrickson's[2] original analysis model, not to conduct a comprehensive methods bake-off. A wider survey belongs to future work (Section 4.6).

Blocks S7 through S9 (Section 3.7) revisit this scoping decision once, for a specific reason. Section 4.1 shows that no carryover term tested so far can repair the interaction's attenuation, no matter how it enters, as long as it enters only as a nuisance main effect. The natural next model interacts the carryover term with the biomarker directly. An earlier version of Block S7 tested two such specifications, `E5` (a lagged term crossed with `bm`) and `E6` (a continuous elapsed-time term crossed with `bm`). Both underperformed Unadjusted at every cell examined across three blocks, two trial designs, two architectures, and three carryover half-lives (Sections 3.6, 4.5), traced to a structural cause: the data-generating process examined throughout this manuscript has a single true decay parameter, so crossing a second carryover term with the biomarker adds a free parameter with no counterpart in the truth being estimated, and two collinear regressors chasing one true degree of freedom lose power to a correctly specified single-parameter model almost by construction.

We expect a further variant of the same interacted-term family to fail on that same structural argument, so rather than test one we replaced `E5`/`E6` with two specifications that answer different open questions about the same underlying problem, the analyst's unknown true decay half-life. `G4` asks whether estimating the half-life from the data recovers the power a correctly assumed half-life would give. It selects the half-life by AIC over a candidate grid, reusing the logic of `characterize_carryover()` (`R/carryover_analysis.R`), already used for real-data analysis in this package. This addresses Block S2's question from the opposite direction: instead of asking how much a wrong assumption costs, we ask whether estimation avoids needing the assumption at all. `G5` asks a more basic question raised directly by the wider literature: is a conditional mixed model even necessary? Senn, Julious and Araujo[7] compared four analysis strategies for ordinary (no biomarker interaction) N-of-1 trials. They found that a simple paired t-test on within-patient treatment differences achieved near-nominal Type I error, higher power, and comparable bias and MSE relative to a carryover-adjusted mixed model, and recommended the mixed model only when carryover is specifically suspected. `G5` is a direct biomarker-interaction extension of their preferred method: a two-stage, correlation-structure-agnostic regression of each patient's own mean on-drug-minus-off-drug difference on the biomarker across patients, with no repeated-measures model and no carryover representation at all.

This remains a bounded comparison. `G4` and `G5` were chosen because each is implementable within, or trivially adjacent to, the existing analysis pipeline (Section 2.7) and each answers a specific question this manuscript's own results raised, not because they close the wider literature gap identified above. The Bayesian, GEE-as-primary-model, randomization-test, and two-stage meta-analytic alternatives remain untested and are discussed as future work (Section 4.6).

2.6 Factorial grid and parameter settings

The principal comparison crosses biomarker moderation, carryover half-life, trial design, and sample size against the three analysis specifications, under covariance moderation and the exponential DGP.

Parameter	Values
Biomarker moderation (c_{bm})	0.0, 0.30, 0.45
Carryover half-life ($t_{1/2}$)	0.0, 0.5, 1.0 weeks
Sample size (N)	35, 70
Trial design	CO, N-of-1 (Hybrid), OL+BDC
DGP architecture	B (covariance moderation) only
Replicates per cell	500 (target), 50 for development

Three specifications, three half-lives, three designs, two sample sizes, and three biomarker levels give $3 \times 3 \times 3 \times 2 \times 3 = 162$ cells. The $c_{bm} = 0$ row is the Type I error check against nominal 5%, and the $t_{1/2} = 0$ row, with no carryover, gives a best-case power reference at each combination of N, design, and c_{bm} against which the cost of nonzero carryover can be read off directly.

Decay-shape sensitivity (Figures 3 and 5) is evaluated separately, at the reference cell only ($c_{bm} = 0.45$, $t_{1/2} = 1.0$ weeks), rather than across the full grid above: exponential decay and two Weibull shapes ($k \in \{0.5, 2.0\}$, Section 2.3) crossed with the three specifications, three designs, and two sample sizes give $3 \times 3 \times 3 \times 2 = 54$ further cells (`23-run-decay-shape-sensitivity.R`). We restrict the decay-shape axis to this one cell because it is the axis every other result in this manuscript holds fixed at exponential; a full crossing of decay shape with the grid above would triple the simulation cost for a comparison this manuscript uses in only two figures. $k = 1.0$ is not a third Weibull condition here, or anywhere in this manuscript: it recovers the exponential form exactly (Section 2.3) and so is not a distinct decay-shape condition.

2.7 Inference and the cluster-robust correction

The interaction test described so far is model-based. The standard error of $\beta_{bm:X}$ is read from the fitted `lme` object, valid only if the assumed residual covariance is correct. That assumption is not innocuous here. The working covariance combines a patient random intercept with a `corCAR1` AR(1) residual correlation, and the companion calibration study[9] shows this does not match the covariance the data-generating mechanism actually induces, inflating the standard error by roughly six to ten percent and depressing the rejection rate correspondingly. We quantify this with the calibration ratio κ, the mean estimated SE divided by the empirical SD of the interaction estimate across replicates ($\kappa = 1$ is correctly calibrated, $\kappa > 1$ conservative). At the null cell κ runs 1.05 to 1.10 across specifications, with empirical Type I error 0.029 to 0.039 against nominal 0.05.

The remedy is a cluster-robust (sandwich) variance estimator, which derives the variance of $\hat{\beta}_{bm:X}$ from observed between-cluster variation rather than the assumed covariance, and so remains valid under mis-specification. Clusters are patients (eight occasions each, unique across randomization paths). Because the naive sandwich form is markedly downward-biased with few clusters ($N = 35$ or 70), we use the CR2 bias-reduced estimator of Bell and McCaffrey[8], paired with Satterthwaite degrees of freedom (both via `clubSandwich`, replacing only the standard error, reference distribution, and p-value, leaving point estimates untouched).

We apply the correction in two places. A null-cell diagnostic ($c_{bm} = 0$, exponential DGP, $t_{1/2} = 1.0$ weeks, Hybrid, $N = 70$, 3,000 replicates) establishes whether CR2 restores nominal size (Section 3.6). A five-cell block, the reference cell plus four stress cells (small N, high autocorrelation, half-life mis-specification, 30% MCAR dropout, 500 replicates each, labeled S6 in this manuscript's block numbering), then asks whether restoring size disturbs the ranking (same section). Blocks S1 through S4 are the Tier 2 sensitivity analyses of Section 3.7. Block S5, an exploratory autocorrelation-by-carryover interaction, is outside the present scope and reported only in the archived full-scope drafts.

2.8 Performance measures and computing environment

For each cell we report Power, Type I error, Bias, Empirical SE (calibrated scale), Coverage of the nominal 95% Wald CI, MSE, non-convergence rate, and the MCSE of each, following Morris et al.[1, Tables 5-6]. We chose $n_{sim} = 500$ so the MCSE on power at $p = 0.5$ would not exceed 0.022. No simulation runs for this manuscript. We filter the same 540-cell production

Table 2: Headline performance at the reference cell (exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $n_{sim} = 500$). Power is the rejection rate of $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$. Bias is on the calibrated effect-size scale, $\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR} = -3.6$ (per docs/19 calibration). MCSEs are reported per Morris et al. (2019, Table 6).

t1/2	Specification	Power (MCSE)	Bias (MCSE)	Empirical SE	Coverage (MCSE)	MSE (MCSE)	Non-conv
0.0	Unadjusted	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	Lag-adjusted	0.990 (0.004)	-0.025 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	Exposure-weighted	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.5	Unadjusted	0.918 (0.012)	+0.586 (0.036)	0.807	0.938 (0.011)	0.995 (0.058)	0.000
0.5	Lag-adjusted	0.920 (0.012)	+0.583 (0.036)	0.812	0.938 (0.011)	1.000 (0.059)	0.000
0.5	Exposure-weighted	0.948 (0.010)	+0.072 (0.041)	0.907	0.972 (0.007)	0.828 (0.057)	0.000
1.0	Unadjusted	0.766 (0.019)	+1.093 (0.038)	0.847	0.828 (0.017)	1.911 (0.093)	0.000
1.0	Lag-adjusted	0.770 (0.019)	+1.099 (0.038)	0.853	0.818 (0.017)	1.935 (0.094)	0.000
1.0	Exposure-weighted	0.860 (0.016)	-0.081 (0.049)	1.085	0.968 (0.008)	1.184 (0.077)	0.000

grid summary used in the archived full-scope drafts down to the 216-cell slice defined by covariance moderation and the two non-linear decay forms. All three specifications are fit to the same simulated dataset within each cell (common random numbers), so every pairwise contrast, especially Lag-adjusted minus Unadjusted, carries reduced Monte Carlo variance relative to independent draws. The Tier 2 sensitivity blocks and S6 recalibration are filtered from the same shared production outputs. Exact file paths and versions are given in the Reproducibility section.

3 Results

All results reported below derive from the same 500-replicate production run used in the full-scope manuscripts, filtered to the covariance-moderation architecture and the exponential and Weibull decay forms. The Monte Carlo standard error on power does not exceed 0.022 at $p = 0.5$, and differences smaller than this should not be over-interpreted. Coverage of the nominal 95% Wald confidence interval is reported for the reference cell, and non-convergence rates are reported per cell in the tables that follow.

3.1 Headline performance table

Table 2 reports performance at the reference cell (exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$), where non-convergence was zero throughout. The bias, MSE, and coverage columns are computed against Exposure-weighted's calibrated target, $\theta_{\text{true}} = -c_{bm}\sigma_{BR} = -3.6$, while Unadjusted and Lag-adjusted estimate a different coefficient, bm:D_{it}. Their apparent bias and under-coverage reflect that estimand mismatch, not a defect in the estimators, so those columns are comparable within the Unadjusted/Lag-adjusted pair but not across to Exposure-weighted. For the three-way comparison we rely on power and Type I error, properties of the test rather than the coefficient tested.

3.2 Tier 1 grid overview

Figures 2 through 4 give a full-grid view of the Tier 1 results as annotated heatmaps, with trial design as column facets, N as row-block facets, analysis specification as the row axis within each block, and a blue (low power) to red (high power) fill scale. Each panel fixes two of the grid's remaining dimensions and varies a third, since no single two-dimensional heatmap can display all four simultaneously. Panels A and B report all nine G1-G9 specifications (a dedicated G1-G9 simulation restricted to the exponential DGP and Architecture B, $n_{sim} = 100$, preliminary); Panel C retains the original three (G1-G3), since it is the only panel that varies DGP decay form, an axis the G1-G9 extension did not re-run.

Figure 2: Tier 1 grid, biomarker-effect axis, all nine G1-G9 specifications: power by trial design (columns), N (row blocks), and analysis specification (rows within block), across $c_{bm} \in \{0, 0.30, 0.45\}$ at $t_{1/2} = 1.0$ weeks, exponential DGP, Architecture B (preliminary $n_{sim} = 100$).

Two features are visible at a glance in Panel A that the headline table makes only cell by cell. First, within every design by N block, `G1` and `G2` (Unadjusted, Lag-adjusted) track each other closely at every biomarker level, differing by at most 0.03, within Monte Carlo noise at this preliminary $n_{sim} = 100$. This matches the higher-precision result of Section 3.4, where the two are numerically identical to two decimal places. Second, the Exposure-weighted family (`G3`, `G4`, and their CR2 variants `G8`, `G9`) visibly separates from `G1`/`G2`/`G6`/`G7` only in the Hybrid and OL+BDC columns. In the CO column the whole family underperforms at every biomarker level past zero. At the reference cell ($c_{bm} = 0.45$, $t_{1/2} = 1.0$), for example, `G4` reaches 0.48 against 0.86 to 0.89 for `G1`/`G2`/`G6`/`G7`. This is the design-dependence of Section

3.4, read directly off the grid now across all nine specifications rather than three. Whether that separation depends on carryover being present at all, Panel A alone cannot show, since it fixes $t_{1/2} = 1.0$ throughout.

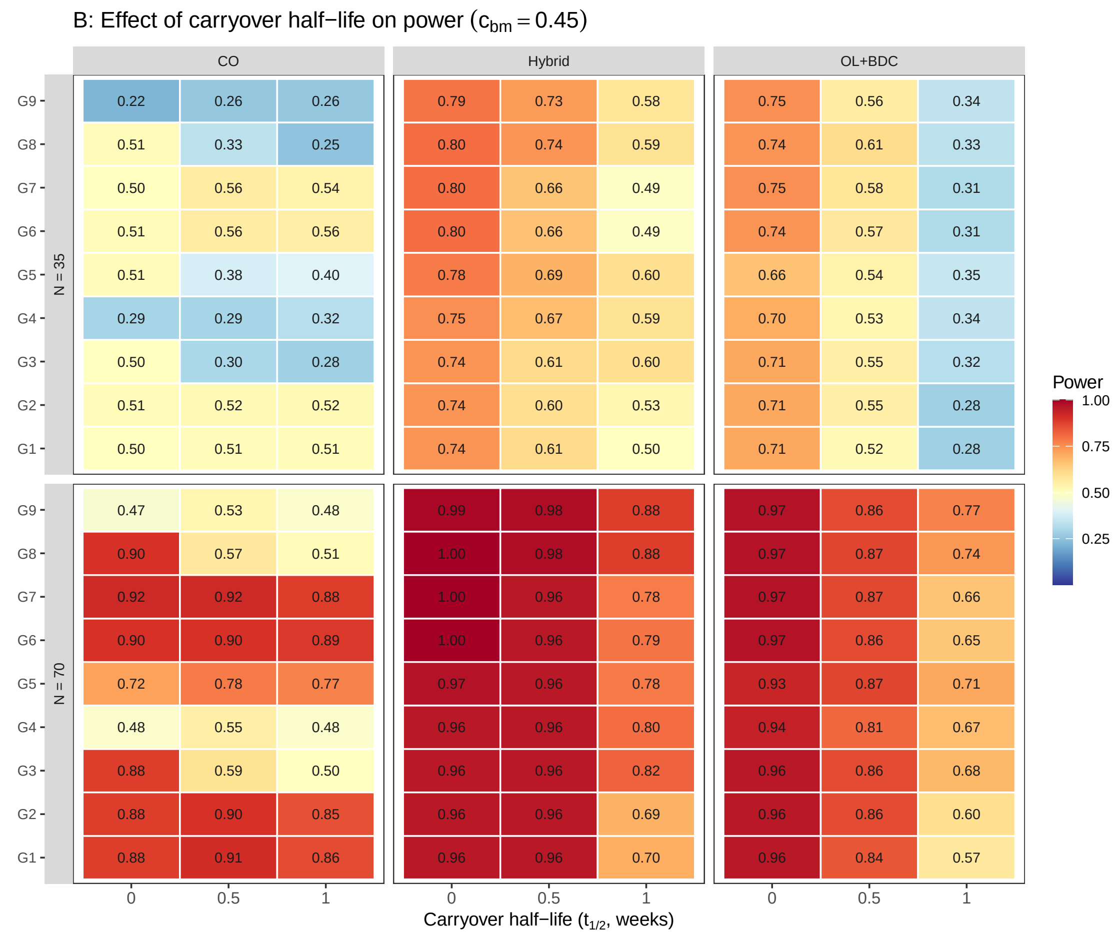

Figure 3: Tier 1 grid, carryover-axis, all nine G1-G9 specifications: power by trial design (columns), N (row blocks), and analysis specification (rows within block), across $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks at $c_{bm} = 0.45$, exponential DGP, Architecture B (preliminary $n_{sim} = 100$).

Panel B answers that, for Hybrid and OL+BDC. At $t_{1/2} = 0$ the Exposure-weighted family's separation from `G1`/`G2`/`G6`/`G7` disappears in both designs (all nine specifications sit at 0.94-1.00), and only opens up as the half-life increases, confirming the separation in Panel A is specifically a carryover effect there (the $t_{1/2} = 1.0$ column is shared with Panel A's reference cell). Under CO the picture is different: `G4` and `G9` are already collapsed at $t_{1/2} = 0$ (0.48 and 0.47, against 0.88-0.92 for the rest), before carryover enters the picture at all. This is the AIC mis-selection pathology of Block S9 read directly off the grid, not a carryover effect: `G4`'s half-life selection drifts to the largest candidate under CO's sparse off-drug sampling regardless of the true half-life, so its CO weakness predates $t_{1/2} > 0$ rather than growing from it. Panel B also adds a ranking result Panel A alone does not show. `G9` and `G8` lead, in a near-tie, at the Hybrid and OL+BDC reference cells (both around 0.88 at Hybrid, 0.77 to 0.74 at

OL+BDC). `G5` (Paired-difference) is the strongest specification at CO that is not `G1`/`G2` or a CR2 variant of them (0.77, versus 0.48 to 0.51 for the Exposure-weighted family there). This is consistent with `G5` discarding the carryover representation that costs the Exposure-weighted family power under CO.

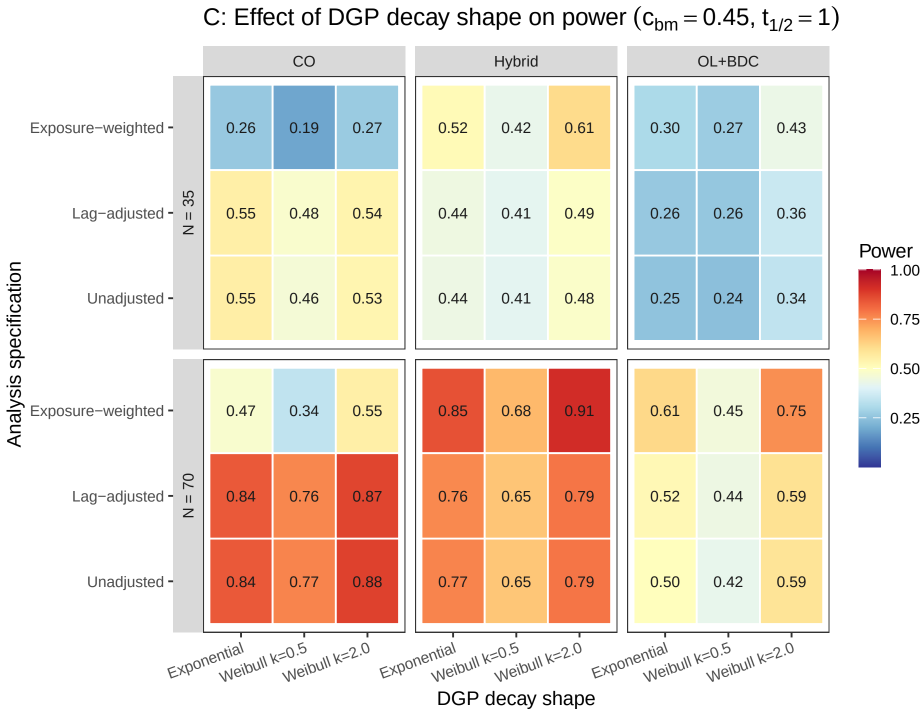

Figure 4: Tier 1 grid, decay-shape axis: power by trial design (columns), N (row blocks), and analysis specification (rows within block), across the three DGP decay-shape levels (exponential, Weibull $k \in \{0.5, 2.0\}$) at $c_{bm} = 0.45$, $t_{1/2} = 1.0$ weeks.

Panel C shows the Exposure-weighted row's advantage in the Hybrid and OL+BDC columns holding across every decay shape examined, including the heaviest-tailed Weibull form ($k = 0.5$), where the advantage narrows most but does not reverse. This is the decay-form robustness of Section 3.5, shown here across the full grid rather than at one cell.

3.3 Matched-decay baseline

At the matched cell (exponential DGP, exposure-weighted analysis, $N = 70$, $c_{bm} = 0.45$, Hybrid design), power declines from 0.988 ± 0.005 at $t_{1/2} = 0$ to 0.948 ± 0.010 at $t_{1/2} = 0.5$ to 0.860 ± 0.016 at $t_{1/2} = 1.0$ weeks (point estimate plus or minus MCSE, $n_{sim} = 500$). All 500 replicates converged at every half-life in this cell.

We had previously read the value at $t_{1/2} = 1.0$ as agreeing with the power of 0.82 reported by Hendrickson et al. for a comparable cell[2]. Unfortunately, we must withdraw that comparison. In the reference implementation, the analysis-side and data-generating carryover parameters are set independently. The drivers that reproduce the original figures supply a half-life to the data-generating step only, leaving the analysis-side half-life at its default of zero. At a zero half-life, the exposure-decayed predictor collapses onto the binary on-drug indicator. The reference analysis is therefore Unadjusted, not Exposure-weighted, so the appropriate comparator is our unadjusted power at the same cell, 0.766. The grids differ further on interaction strength and half-life, so ‘the same cell’ was approximate in any case. We have not resolved what the published figure was computed under, and make no claim of agreement or disagreement here. Nothing else in this manuscript depends on that comparison, since the analysis specifications are compared against one another within a single simulation under common random numbers.

3.4 Power by analysis specification

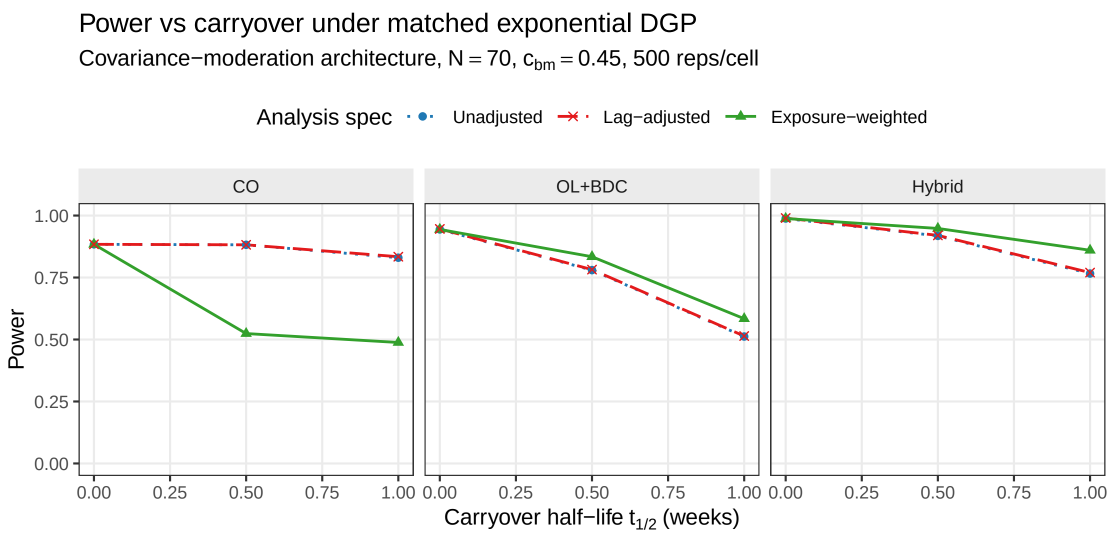

Figure 5: Power versus carryover half-life under the exponential DGP, by analysis specification and trial design.

Two features of Figure 5 carry the substance of this study. First, the Unadjusted and Lag-adjusted curves lie on top of one another in every panel and at every half-life. At the reference cell they give 0.766 and 0.770 at $t_{1/2} = 1.0$ weeks, and across all 216 cells the mean difference is 0.003, with mean bias, MSE, and coverage agreeing to three decimal places. Since the two share an estimand, are fitted to the same simulated datasets, and differ by exactly one nuisance term, this is as clean a null as the design can produce. The lagged-treatment term contributes nothing on any measure (Section 4.1). Second, the three specifications give nearly identical power at $t_{1/2} = 0$, where there is no carryover to model. They diverge as the half-life increases, with Exposure-weighted separating from the other two. It retains a modest but consistent advantage in the Hybrid and OL+BDC designs from $t_{1/2} = 0.5$ weeks onward. Under the crossover design, though, it is markedly inferior to both alternatives (0.488 against 0.830 for Unadjusted). We think this happens because the within-subject correlation in a crossover trial already carries most of the signal the decaying predictor targets. It gains little there, while its down-weighting of off-drug observations works against it (Section 4).

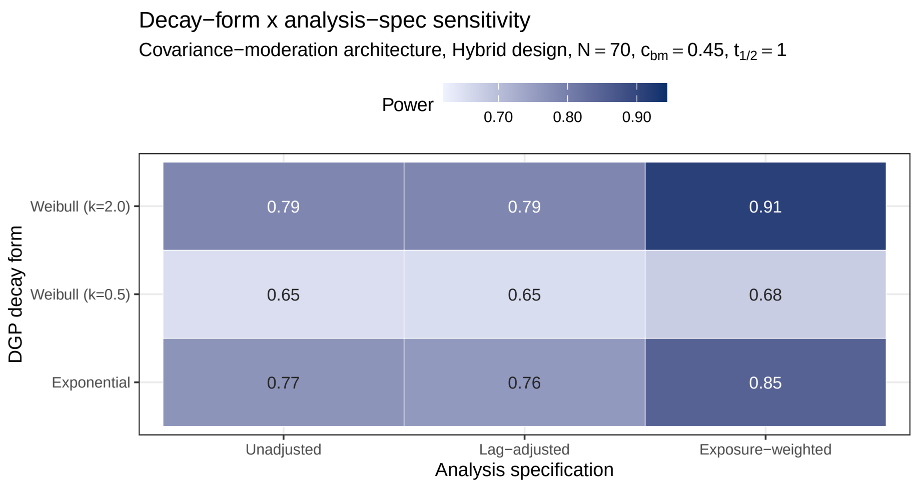

Figure 6: Power at the critical cell ($t_{1/2} = 1.0$ weeks, Hybrid, $N = 70$, $c_{bm} = 0.45$) across DGP decay forms (rows) and specifications (columns). Fill scale is stretched to the observed range, not [0,1], and not comparable to other figures.

3.5 Decay-form mis-specification

The heatmap cell at the intersection of the exponential DGP row and the Exposure-weighted column is the matched reference, with power 0.850 ± 0.016 in this figure's own simulation (a dedicated decay-shape run, Section 2.6; the 0.860 reported for the same nominal cell in Table 2 and Figure 5 comes from an independent draw of the same DGP, and the two agree within one MCSE). The Unadjusted and Lag-adjusted columns, invariant to the assumed decay form by construction, are near-identical throughout, as in Figure 5, so variation down those columns is data-generating decay and Monte Carlo noise. The remaining Exposure-weighted rows give the cost of assuming exponential decay when the truth is Weibull. Even at the heaviest-tailed form examined ($k = 0.5$), where the advantage narrows most (power 0.678 against 0.652 and 0.654 for Unadjusted and Lag-adjusted), it does not reverse (Section 4).

3.6 Type I error control and cluster-robust recalibration

Empirical Type I error at the $c_{bm} = 0$ cells of Figure 2 sits uniformly below the nominal 0.05 level for every one of the nine specifications, across every design and N examined there. This is not sampling noise but the conservatism of the model-based test (Section 2.7), from a working covariance that does not match the data-generating one. Because the conservatism is uniform across specifications, it does not disturb the comparison among them, and the cluster-robust recalibration below confirms that a CR2 standard error restores nominal rejection without changing which specification leads. This also establishes that the null result of Section 3.4 is not an artifact of differential test size, since Unadjusted and Lag-adjusted are conservative to the same degree. Table 3 makes this explicit at the null cell, at a higher replicate count (3,000) than the Tier 1 grid affords.

Table 4 confirms this directly. The model-based rejection rate sits below nominal across all five S6 cells and all three specifications, while CR2 restores it to within the Monte Carlo

Table 3: Conservatism of the model-based interaction test and its cluster-robust correction, at the null cell ($c_{bm} = 0$, exponential DGP, $t_{1/2} = 1.0$ weeks, Hybrid design, $N = 70$, 3,000 replicates). κ is the calibration ratio defined in Section 2.7. $\kappa > 1$ indicates a conservative test. All three specifications are conservative under the model-based test, most so Exposure-weighted ($\kappa = 1.10$, Type I = 0.029). The CR2 bias-reduced cluster-robust standard error restores κ to near one and Type I to nominal (0.045 to 0.049) for all three.

Specification	κ model	Type I model	κ CR2	Type I CR2
Unadjusted	1.06	0.036	0.98	0.048
Lag-adjusted	1.05	0.039	0.98	0.049
Exposure-weighted	1.10	0.029	1.00	0.045

Table 4: Cluster-robust recalibration at the reference cell and four stress cells (500 replicates, $c_{bm} = 0.45$). EW is Exposure-weighted, LA is Lag-adjusted, U is Unadjusted. κ is the calibration ratio of Section 2.7, here for Exposure-weighted. A value above one is a conservative test. The two gap pairs give each alternative against the unadjusted benchmark under each inference. CR2 df is the mean Satterthwaite degrees of freedom for the cluster-robust test on Exposure-weighted. Computed with `clubSandwich`.

Cell	κ model	κ CR2	EW model	EW CR2	EW−U model	EW−U CR2	LA−U model	LA−U CR2	CR2 df
Reference, $N = 70$	1.15	0.99	0.829	0.863	+0.080	+0.092	-0.003	+0.002	23
Small $N = 35$	1.13	0.96	0.558	0.590	+0.066	+0.086	-0.002	+0.000	12
High autocorr, $\rho = 0.9$	1.26	0.94	0.954	0.980	+0.052	+0.032	+0.000	-0.004	23
Half-life mis-spec	1.14	1.00	0.759	0.795	+0.010	+0.024	-0.003	+0.002	23
30% MCAR dropout	0.99	0.85	0.148	0.126	+0.010	+0.010	-0.004	+0.010	5

band around 0.05. In the information-rich cells the calibration ratio κ runs from 1.07 to 1.26 under the model-based test, restored to about one under CR2, recovering two to five points of power. The table reports both alternatives against the unadjusted benchmark. The Exposure-weighted advantage is held or slightly widened under CR2 at the reference cell (+0.080 to +0.092), the small-N cell (+0.066 to +0.086), and the half-life mis-specification cell (+0.010 to +0.024). It narrows under high autocorrelation (+0.052 to +0.032, though it still leads), and is small under 30% MCAR dropout (+0.010), which we read as the S3 finding restated, since dropout removes the off-drug information the advantage depends on. The Lag-adjusted gap, by contrast, is within ± 0.010 of zero at every cell under either inference, corroborating the Section 3.4 null under a second procedure and at four stress cells the main grid does not cover. This recalibration was validated only in the Hybrid, $N = 70$ cells, and should not be extended to the crossover design, where Exposure-weighted does not lead in the first place.

3.7 Sensitivity analyses

Nine Tier 2 blocks (S1-S4, S7, S8, S9, S10, S11) perturb one axis at a time around a common reference cell (Hybrid, $N = 70$, $c_{bm} = 0.45$, exponential DGP, $t_{1/2} = 1.0$), holding the remaining factors fixed. Two exceptions are S9, which changes the trial design itself, and S11, which sets $c_{bm} = 0$. A tenth block, S6, applies the cluster-robust correction of Section 3.6 at that same cell plus four stress cells, and an eleventh, S5, is exploratory and reported only in the archived full-scope drafts. The blocks are reported separately here and are not pooled with Tier 1 or with each other. Four blocks (S1, S4, S10, S11) are reported here only as a one-paragraph summary, with full cell-by-cell detail, tables, and figures in the Supplement; two more (S2, S8) keep their headline finding and a summary paragraph here, with the remainder of their detail likewise moved to the Supplement.

3.7.1 S1: within-factor autocorrelation

All nine specifications gain power monotonically with autocorrelation $\rho \in \{0.5, 0.7, 0.8, 0.9\}$. The Exposure-weighted advantage over Unadjusted and Lag-adjusted, roughly 0.05 to 0.08 in absolute power, is preserved at every level examined. This suggests that autocorrelation strength and specification choice act additively rather than interacting. Lag-adjusted tracks Unadjusted throughout (Supplement, Section S1).

3.7.2 S2: analyst-truth half-life mismatch

Block S2 crosses two true half-lives ($t_{1/2} \in \{0.5, 1.0\}$ weeks) with four analyst-assumed values ($\hat{t}_{1/2} \in \{0.25, 0.5, 1.0, 2.0\}$ weeks). The penalty for an incorrect assumption is modest throughout (Exposure-weighted stays within about 7 points of its correctly-specified power at every cell but one), with an asymmetry: under-assuming the half-life is safer than over-assuming it. This follows from the limiting-case relationship of Section 2.4, since shrinking $\hat{t}_{1/2}$ drives the predictor toward the binary indicator, which remains serviceable, whereas inflating it drives the predictor toward a constant, at which the on-off contrast and the estimand itself degenerate. Full cell-by-cell detail, figures, and the nine-specification extension are in the Supplement, Section S2.

3.7.3 S3: dropout

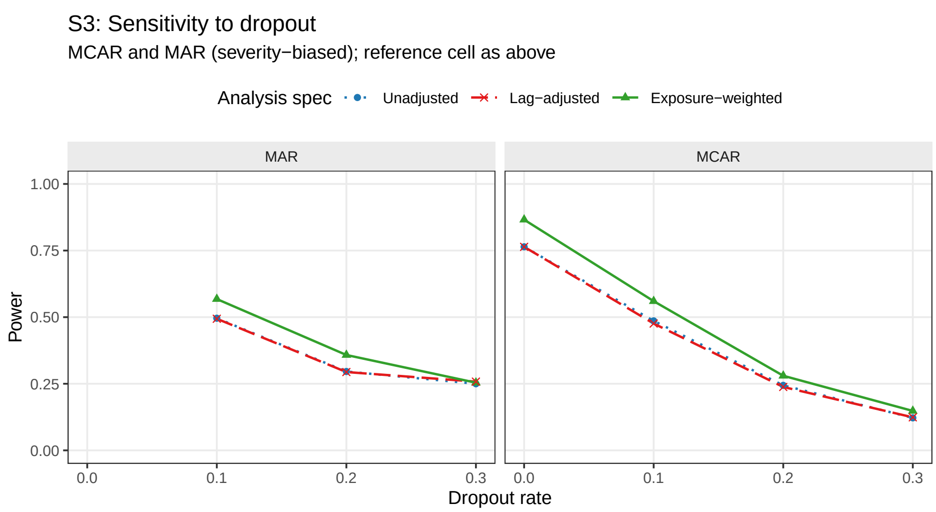

Figure 7: Power at dropout rates $\in \{0, 0.1, 0.2, 0.3\}$ under MCAR and MAR mechanisms.

All three specifications lose power monotonically as dropout increases. At the MCAR baseline ($N = 70$, no dropout), Exposure-weighted gives 0.866±0.015 against 0.764±0.019 for the other two. At 30% MCAR dropout, all three collapse to 0.122 to 0.148. The Exposure-weighted advantage narrows steadily as dropout worsens, from about 0.10 absolute at zero dropout to

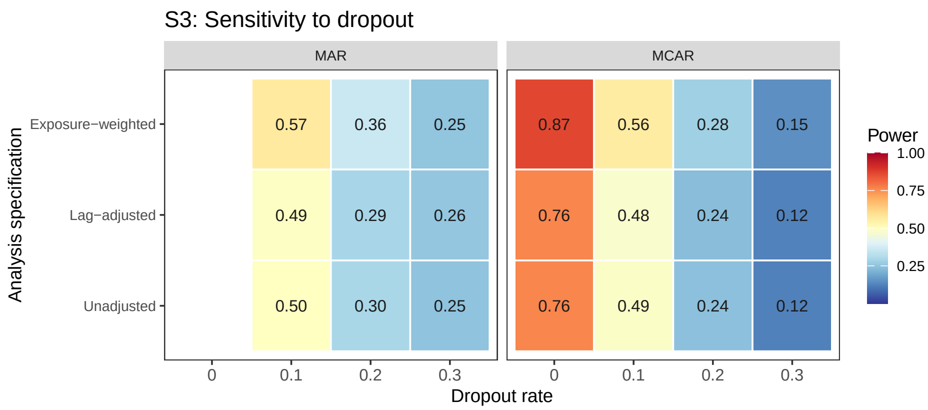

Figure 8: Block S3: power by dropout rate and analysis specification, faceted by dropout mechanism, same data as the line plot above. The MAR arm has no 0% cell (Section 3.7); it shares the MCAR reference cell.

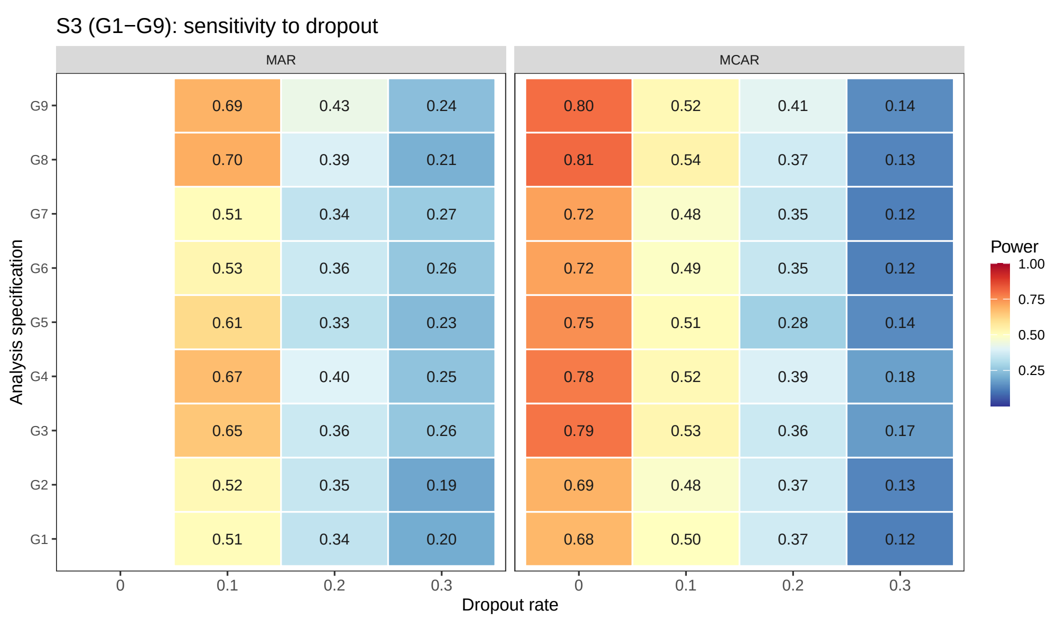

Figure 9: Block S3 extended to all nine G1-G9 specifications (preliminary $n_{sim} = 100$).

Table 5: Block S7: all five G1-G5 specifications at the reference cell (Hybrid, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$, exponential DGP, $n_{sim} = 500$, all replicates converged). G4 targets bm:Dbc at the AIC-selected half-life (candidate grid 0.25, 0.5, 1.0, 2.0); G5 targets the slope of the patient-level diff bm regression. Neither is the same estimand as bm:Db (G1/G2), so only power is comparable across rows. Mean selected t1/2 is the average AIC-chosen half-life for G4 across replicates, against a true half-life of 1.0.

Specification	Power (MCSE)	Mean selected t1/2
G1: Unadjusted	0.748 (0.019)	–
G2: Lag-adjusted	0.748 (0.019)	–
G3: Exposure-weighted	0.826 (0.017)	–
G4: AIC-selected t1/2	0.832 (0.017)	1.31
G5: Paired-difference	0.804 (0.018)	–

0.07 at 10%, 0.04 at 20%, and 0.03 at 30% under MCAR, narrowing further under MAR at the highest rate. This is consistent with the advantage originating in the exposure-weighted contrast from off-drug observations, which contracts as those observations are preferentially lost. The advantage is preserved at dropout of 20% or below. Lag-adjusted again tracks Unadjusted throughout, the largest discrepancy being 0.010.

3.7.4 S4: biomarker-moderation effect-size curve

The power curve is sigmoidal for all three original specifications across $c_{bm} \in \{0, 0.10, 0.20, 0.30, 0.45, 0.60\}$, saturating once $c_{bm} \geq 0.45$. The Exposure-weighted advantage over the unadjusted benchmark is largest at intermediate effect sizes (+0.10 absolute at $c_{bm} = 0.45$), shrinking as all three approach the ceiling, so the advantage matters most for trials targeting a moderate biomarker-treatment interaction (Supplement, Section S4, full curve and nine-specification extension).

3.7.5 S7: alternative-paradigm specifications

An earlier version of this block tested two carryover terms crossed with the biomarker, `E5` (Sx $\sim bm + t + D_{it} + \text{bm:}D_{it} + L_{it} + \text{bm:}L_{it}$) and `E6` (Sx $\sim bm + t + D_{it} + \text{bm:}D_{it} + t_{sd} + \text{bm:}t_{sd}$). This was motivated by Section 4.1's observation that a carryover term entering only as a nuisance main effect cannot repair the interaction's attenuation. Both underperformed Unadjusted at every cell examined, evaluated by a joint 2-df Wald test since each adds a second interaction coefficient. We traced this to a structural cause discussed in Section 4.6: this manuscript's DGP has a single true decay parameter, so a second, DGP-unmotivated interaction coefficient mostly adds estimation noise. `E5`/`E6` were replaced by two specifications addressing the same open problem, the analyst's unknown decay half-life, from different angles (Section 2.4): `G4` (AIC-selected half-life) and `G5` (paired-difference regression). Both are single-coefficient tests, unlike `E5`/`E6`.

Table 5 gives a genuinely different result from the retired `E5`/`E6` comparison. Both `G4` and `G5` beat Unadjusted at this cell, and `G4` edges out `G3` (Exposure-weighted) as well. `G4` leads overall (0.832), `G3` follows (0.826), then `G5` (0.804, the two-stage paired-difference regression

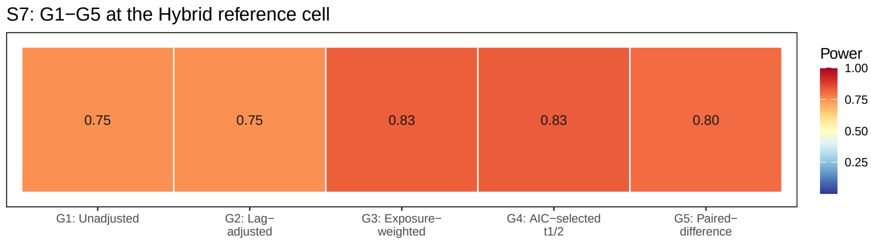

Figure 10: Block S7: power by analysis specification, same figures as the table above.

with no repeated-measures model at all), and `G1`/`G2` tie at the back (0.748 each, the already-established Section 3.4 null replicated once more).

`G4`'s AIC selection is imperfect but usefully informative. The mean selected half-life is 1.31 weeks against a true value of 1.0, and the selection distribution concentrates on the true value and its immediate neighbor (88% of replicates choose 1.0 or 2.0 weeks). This is close enough to the truth that `G4` slightly exceeds `G3`'s own oracle-half-life power at this cell.

We see a more notable result in `G5`. It discards the repeated-measures structure, the correlation model, and any carryover representation entirely, and still beats `G1`/`G2`. This echoes Senn, Julious and Araujo's[7] finding that simplicity is not obviously costly for N-of-1 treatment-effect estimation. None of these results should be read as general, however. Block S9, next, tests whether the ranking survives a different trial design, and it does not for every specification equally.

3.7.6 S8: architecture sensitivity of the specification ranking

Every result reported to this point, Tier 1 and blocks S1-S7 alike, is restricted to Architecture B (covariance moderation), the manuscript's stated scope (Section 2.3). Block S8 asks a narrower question than a full architecture comparison, which is the purpose of the companion manuscript on DGP architectures[3]: does the ranking among analysis specifications established above under Architecture B hold under Architecture A (mean moderation) as well, or is the choice of carryover representation itself architecture-dependent? We refit the five specifications of block S7's table at the same reference cell under both architectures, at three half-lives ($t_{1/2} \in \{0, 0.5, 1.0\}$).

The S7 result (`G4` and `G5` both beating `G1`) turns out to be architecture-specific, joining Exposure-weighted in a pattern that grows with carryover severity rather than switching on at some threshold. Under Architecture B, `G4` leads at $t_{1/2} = 1.0$ (0.850), ahead of `G5` (0.842), `G3` (0.836), and `G1`/`G2` (0.760). Under Architecture A the pattern reverses for `G4` and `G3` alike: `G1`/`G2` lead (0.972/0.976), `G5` trails only slightly (0.964), while `G3` (0.946) and `G4` (0.950) trail by 2 to 3 points, a real if modest reversal. This matches the archived full-scope drafts' finding that Exposure-weighted is marginally dominated under mean moderation (Section 4.6). `G4` inherits the same sensitivity, since it shares Exposure-weighted's formula. `G5` is comparatively more robust to architecture, never reversing to a large disadvantage either way. Lag-adjusted (`G2`) remains statistically indistinguishable from Unadjusted (`G1`) under both architectures at every half-life, extending Section 3.4's clean null to Architecture A as well. The one specification recommendation this manuscript makes, exposure weighting in the Hybrid and OL+BDC designs (Conclusions), is therefore itself conditional on Architecture B and grows more important as carryover worsens. The same qualification now extends to `G4`. Full cell-by-cell detail, the table, and both architecture panels are in the Supplement, Section S8.

3.7.7 S9: does the reference-cell ranking survive under CO?

Blocks S7 and S8 found `G4` and `G5` both beat Unadjusted at the Hybrid reference cell under Architecture B, with `G4` slightly exceeding Exposure-weighted (`G3`) itself. Block S9 asks whether that ranking is specific to Hybrid's short, compressed washout, by refitting all five G1-G5 specifications under the CO design (off-drug arm spanning 2.5 to 20 weeks post-discontinuation, Section 2.2) at two true half-lives.

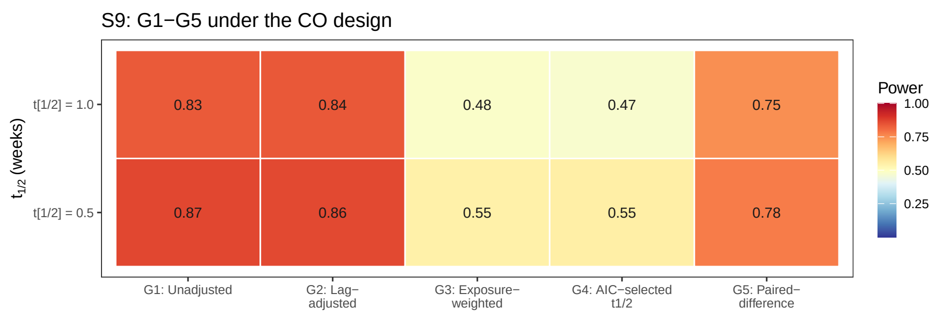

Figure 11: Block S9: power under the CO design at two true half-lives, all five G1-G5 specifications.

It is not, and including `G3` in this block changes the diagnosis reached from `G4` alone. At $t_{1/2} = 1.0$, `G1`/`G2` lead (0.834/0.838); `G5` trails moderately (0.750); `G3` and `G4` both collapse, to 0.482 and 0.474 respectively, statistically indistinguishable from each other (gap 0.008, well under one MCSE). At $t_{1/2} = 0.5$ the same pattern holds (0.866/0.864 for `G1`/`G2`, 0.778 for `G5`, 0.546/0.552 for `G3`/`G4`, again indistinguishable). Because `G3` uses the *true* half-life, not an estimated one, its near-identical collapse shows that `G4`'s failure under CO is inherited almost entirely from Exposure-weighted's own established weakness under this design (Section 3.4: 0.488 versus 0.830 in the Tier 1 grid), not created by the AIC-selection step. The selection step does have a real, separately confirmed pathology under CO: it concentrates overwhelmingly on the largest candidate half-life in the grid (2.0 weeks, chosen in 483 of 500 replicates at $t_{1/2} = 1.0$ and 435 of 500 at $t_{1/2} = 0.5$, regardless of which true half-life generated the data), consistent with a weakly informative likelihood surface across candidates when off-drug occasions are few and widely spaced. But that mis-selection costs `G4` essentially nothing beyond what `G3` already loses under CO with the correct half-life in hand. The dominant story is the continuous decayed predictor's mismatch with CO's design generally (Section 3.4). AIC selection is an identified but practically minor addition to that story here, not the primary cause it appeared to be from `G4` in isolation.

`G5`'s milder reversal has no comparable mechanical explanation. It simply reflects that CO's own within-subject contrast (Section 3.4) already captures most of the signal a between-patient paired-difference regression is built to extract, so discarding the repeated-measures structure costs more there than it does under Hybrid. None of `G3`, `G4`, or `G5`'s Hybrid-cell behavior is design-general, then. The design-specific guidance in Section 4.5 ('What analysis is best in which situations, in practice') should be read accordingly: `G3`/`G4` should not be used under CO regardless of whether the half-life is assumed or estimated.

3.7.8 S10: does a CR2 correction change the ranking among G1-G9?

Section 3.6 (Block S6) showed the model-based interaction test is mildly conservative, and that a CR2 cluster-robust standard error recovers two to five points of power without disturbing the ranking among `G1`, `G3`, and `G2`. That correction had not been applied to `G4` or `G5`. This block asks whether CR2 helps `G4` and `G1`/`G2` the way it already helps `G3`, and whether it changes which specification leads at the Hybrid, Architecture B reference cell. We use the compact `G1`-`G9` naming layer for the resulting nine-way comparison: `G6`-`G9` cross CR2 with `G1`, `G2`, `G4`, and `G3`. `G5` has no CR2 variant, since its paired-difference regression collapses to one row per patient before fitting, leaving no repeated-measures cluster structure for CR2 to correct.

CR2 helps every specification it was applied to: `G1`/`G2` (Unadjusted and Lag-adjusted, both 0.748 model-based) rise to `G6` 0.776 and `G7` 0.764. The ranking among the five base specifications is unchanged by these gains, since `G1`/`G2` remain well below `G3`-`G5` regardless. What does change the ranking is `G9`: CR2 lifts `G4` (AIC-selected half-life, 0.832 model-based) to 0.870, a 3.8-point gain, the largest CR2 correction observed anywhere in this manuscript, enough for `G9` to overtake every other specification tested, including `G8` (0.863). This did not come with a Type I error cost (Section S11 of the Supplement checks this directly): every replicate converged under `G9`, and the power gain is at least consistent with CR2 correcting real conservatism rather than papering over a different problem. Full detail, the reference- cell table, and the heatmap are in the Supplement, Section S10.

3.7.9 S11: is G4's Type I error controlled?

`G4` selects its half-life by AIC and then tests the interaction coefficient from the selected model, a post-selection inference problem: the same data drive both the selection and the test, which can in principle inflate Type I error beyond nominal. Block S11 checks this directly, refitting `G1`-`G5` and `G9` at the null ($c_{bm} = 0$) at the Hybrid reference cell, with $n_{sim} = 2000$ for tighter precision on a rate expected near 0.05.

The concern does not materialize. `G4`'s Type I error is 0.038, conservative and in the same range as `G1`-`G3` (0.034-0.036), not inflated by the selection step. `G9` (`G4`+CR2) is 0.052, indistinguishable from nominal 0.05. Read together with Section 3.6's power results, `G9`'s status as the best-performing specification (Section 3.7, Block S10) is not an artifact of an uncontrolled test: it wins on power while holding Type I error at nominal, the same standard `G3`+CR2 (`G8`) is held to. This does not generalize beyond the single cell tested. Full detail and the heatmap are in the Supplement, Section S11.

3.8 A higher-precision check

Because a number of the Tier 1 gaps are small, we reran a focused 24-cell grid (OL+BDC design, $N \in \{35, 70\}$, true half-life $\in \{0.5, 1.0\}$) at 600 trials per cell, all converging. We compared specifications with a paired McNemar test, considerably more sensitive than an independent-groups comparison since all three are fit to the same trials. Type I error across the twelve null cells fell in $[0.003, 0.053]$, so nominal alpha holds. Paired McNemar tests place Exposure-weighted ahead of the unadjusted benchmark at all four alternative cells, by +2.3 to +13.7 percentage points. The largest gap is at the highest-leverage cell ($N = 70$, true $t_{1/2} = 1.0$ weeks, $p = 6.6 \times 10^{-17}$), with $p \leq 0.015$ at the remaining three, a sharper confirmation than the Tier 1 grid alone provides.

One scope caveat governs the rerun's third arm. It was produced by the package's `lme_analysis()` routine, whose carryover option appends time since discontinuation as a linear main effect rather than the lagged indicator L_{it} used in the Tier 1 grid. It is therefore not the Lag-adjusted specification of Section 2.4 and is not reported as such here. That alternative remains a reasonable specification we have not evaluated (Section 4.6).

4 Discussion

4.1 The lagged-treatment term does no work

The clearest result of this study is a negative one. Adding the classical lagged-treatment indicator to the analysis model changed nothing measurable, anywhere in the grid. Across all 216 cells the mean power difference against the unadjusted benchmark was 0.003 against an MCSE of 0.018. Bias, MSE, and coverage agreed to three decimal places, and the equality held in every sensitivity block and all five cluster-robust stress cells under both inference procedures.

The strength of this conclusion rests on the design rather than the grid size. The two specifications estimate the same coefficient, were fitted to the same simulated datasets under common random numbers, and differ by exactly one term. There is therefore no confounding from a change of estimand, from Monte Carlo variation between arms, or from differential test size (both are conservative to the same degree, Section 3.6). What is left is the term itself, and it contributed nothing.

The mechanism follows from where the term sits. Carryover damages the interaction test on two fronts (Section 1). It attenuates the coefficient, because contaminated occasions are coded as drug-free, and it inflates the residual variance, because those occasions are heterogeneous in a way the model does not represent. The lagged indicator enters only as a nuisance main effect, with no bm:L_{it} product term, so the tested interaction remains bm:D_{it} and the attenuation is untouched. It can in principle absorb some of the second effect, but evidently absorbs too little to matter, since it marks a single occasion per randomization path and is constant thereafter, unable to describe a graded decline across a two- to four-occasion washout.

A more capable, elapsed-time-indexed nuisance term might succeed where the crude lagged indicator fails. We therefore ran a targeted check on the alternative the reference implementation provides, `Sx ~ bm + t + Db + bm:Db + tsd`, called the washout-adjusted specification (`E4`), across the three designs at true half-lives of 0.5 and 1.0 weeks ($N = 70$, $c_{bm} = 0.45$, exponential DGP, 500 replicates). It does slightly better, and the improvement is instructive rather than material. Type I error is at nominal for both (mean 0.050 versus 0.045). Power exceeds the benchmark in all six cells by a mean of 1.3 points (sign test $p = 0.031$), reaching significance under crossover at both half-lives (+1.8, $p = 0.008$; +2.2, $p = 0.003$) and OL+BDC at the longer one (+2.2, $p = 0.015$), but not Hybrid (+1.0, $p = 0.27$). The mechanism is exactly the one Section 2.4 predicts. Point estimates are unchanged, so nothing about the attenuation is repaired. The entire effect runs through the standard error, which only a long enough washout gives it something to describe. The crossover schedule places off-drug occasions 2.5 to 10 weeks after discontinuation, against one to two weeks for Hybrid. The practical ceiling on what any carryover nuisance term can deliver is therefore about two percentage points, and only in long-washout designs, against the nine to eleven points exposure weighting delivers where it works. This check covers only the exponential DGP at a single N and effect size, so it is a probe rather than a fourth arm. It shares every specification's structural limitation of entering carryover as a main effect only (Section 4.6). For practitioners the implication is direct. A carryover nuisance term of either form should not be expected to recover meaningful power lost to carryover, and reporting one does not constitute an effective adjustment for it.

4.2 The exposure-weighted advantage, and where it fails

The exposure-weighted predictor does help, but conditionally. At its best cell (Hybrid, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$) it exceeds the unadjusted benchmark by about 10 points of power (0.860 versus 0.766, roughly five MCSEs and so not attributable to chance), confirmed at all four cells of the Section 3.8 high-precision rerun. Under the crossover design it performs considerably worse (0.488 versus 0.830). We think this happens because the within-subject correlation there already carries most of the signal the decaying predictor was designed to recover, so it gains little while its down-weighting of off-drug observations works against it. The recommendation therefore holds only in the two non-crossover designs, and should not be extended to a crossover trial, where the unadjusted binary indicator is materially better.

That the one specification addressing attenuation is also the only one to gain power, while the one addressing only residual variance gains nothing, suggests attenuation is the mechanism that matters here. We read this as an interpretation consistent with the results, not as a finding the design isolates directly.

4.3 What the robustness checks add

The Section 3.7 sensitivity blocks show that the exposure-weighted advantage, where it exists, is not fragile. It survives incorrect decay-shape assumptions (Figure 6) and an incorrectly assumed half-life, short of a large over-assumption at a short true half-life (Block S2, Supplement Section S2). It also holds its five- to eight-point lead across the autocorrelation range examined (Block S1, Supplement Section S1).

One caveat qualifies the Tier 1 figures throughout. Outside block S2, the analysis-side half-life is set equal to the data-generating one. This grants Exposure-weighted an oracle no real analyst possesses, so its Tier 1 margins are upper bounds, and block S2 gives the more realistic figure.

4.4 Dropout matters more than the method

The finding with the largest practical consequence concerns missing data, not carryover. When patients drop out (Figure 7), power collapses for all three specifications together, to between 0.12 and 0.25 at 30% dropout, essentially indistinguishable. No specification offers protection, since each can only organize whatever data remain. MAR dropout is somewhat worse than MCAR, because the sickest patients, who carry the most signal, tend to leave first. Retaining patients in the trial yields considerably more power than any choice among these three specifications ever could.

4.5 What analysis is best in which situations, in practice

Collecting the findings above into a decision rule rather than a set of pairwise comparisons:

- **Covariance moderation (Architecture B), Hybrid or OL+BDC design.** Use Exposure-weighted with a CR2 cluster-robust standard error. The advantage holds under decay-shape mis-specification (Section 3.5), under half-life mis-specification short of a large over-assumption (Section 3.7, Block S2), and across the autocorrelation range examined (Block S1), and CR2 restores nominal Type I error without disturbing the ranking (Block S6, validated at $N \in \{35, 70\}$ in the Hybrid design specifically; OL+BDC was not itself stress-tested under CR2). This remains the recommended default: it has been through the full robustness battery `G4`/`G5` have not (next point).
- **`G4` (AIC-selected half-life) and `G5` (paired-difference) are promising at this same cell, not yet validated defaults.** Both beat Exposure-weighted itself at the Hybrid, Architecture B reference cell (Blocks S7-S8: `G4` 0.850, `G5` 0.842, `G3` 0.836, at $t_{1/2} = 1.0$), and `G4` in particular removes the need to assume a half-life at all. Neither has been tested against decay-shape mis-specification, dropout, or under CR2 the way `G3` has, so this is grounds to investigate them further, not to displace the established default yet. Adding a CR2 correction to `G4` (`G9` in the nine-way naming layer) goes further still: at this same cell it outperforms every one of the other eight specifications, including `G3`+CR2, with no accompanying Type I error inflation (Section 3.7, Blocks S10-S11). This is the single best result anywhere in this manuscript, and also the least validated. It rests on one cell, with none of `G4`'s own robustness gaps yet closed.
- **Crossover (CO) design, any architecture.** Use Unadjusted. Exposure-weighted is markedly worse under CO (Section 3.4), and Block S9 shows `G4` and `G5` fail here too, `G4` catastrophically. `G4`'s AIC half-life selection systematically drifts to the largest candidate in its grid under CO's sparse, widely spaced off-drug sampling (chosen in over 85% of replicates regardless of the true half-life), collapsing its power to roughly half of Unadjusted's. It should specifically be avoided under any design with few, widely spaced off-drug observations. `G5` also underperforms Unadjusted under CO, though far more mildly. The classical lagged-treatment remedy adds nothing here either (next point),

including in the one design where its washout is long enough to give it a chance (Section 4.1).

- **Architecture uncertain, or known to be mean moderation (Architecture A).** Use Unadjusted or Lag-adjusted, not Exposure-weighted or `G4`. Both specifications' advantage under covariance moderation reverses, marginally, under mean moderation (Block S8). `G5` is the one specification whose performance does not depend much on which architecture generated the data, small under B, smaller under A, but it has been tested at only one design and reference cell outside the Hybrid comparison (Block S9).
- **Never rely on Lag-adjusted as a carryover remedy.** It is statistically indistinguishable from Unadjusted everywhere tested in this manuscript, Tier 1 and every Tier 2 block alike (Section 4.1). If a reviewer or protocol expects some carryover term to be present, including it costs nothing measurable, but it should not be reported as having adjusted for carryover.
- **Minimize dropout before optimizing specification choice.** It is the single largest lever in the entire grid (Section 4.4), and no specification, including `G4`/`G5`, has been shown to offer any protection against it; their behavior under dropout is untested.

4.5.1 Why G4 and G5 succeed at Hybrid but fail under CO

Both new specifications' Hybrid, Architecture B advantage evaporates under the CO design (Block S9), for two different, design-specific reasons rather than a shared structural flaw of the kind that sank `E5`/`E6` (Section 4.6). `G4`'s failure is mechanical. With only a few, widely spaced off-drug occasions, the likelihood surface across candidate half-lives is weakly informative, and AIC selection drifts to the boundary of its candidate grid rather than locating the true value, systematically overestimating the half-life. Section 3.7's S2 block already showed that over-assumption is the more damaging direction of half-life error when the half-life is fixed by the analyst. Under CO, `G4` reproduces that same damaging error through the estimation procedure itself, on nearly every replicate.

This is a concrete, testable prediction: `G4` should be reliable whenever a design gives several off-drug occasions across a graded range of elapsed time, as Hybrid does despite its short overall washout, and unreliable whenever off-drug sampling is sparse, however long the washout runs in calendar time.

`G5`'s failure has no comparable mechanical account and is milder. The most plausible explanation is that CO's own within-subject contrast (Section 3.4) already carries most of the signal a between-patient paired-difference regression is built to extract, so discarding the repeated-measures and correlation structure costs more under CO than it does under Hybrid, where that structure is doing more work. This was not tested directly and would require comparing `G5` against the within-cell decomposition of `G3`'s advantage to confirm.

Two implications follow. First, these results reinforce rather than undercut the earlier `E5`/`E6` diagnosis (Section 4.6). The more fundamental unresolved question is not which design an alternative specification is tested under, but whether the underlying data-generating process rewards that specification's added structure at all, and neither `G4` nor `G5` was tested against a DGP built to favor them specifically. Second, the practical recommendation is narrower than ‘estimate the half-life’ or ‘use a simpler method’: both `G4` and `G5` are candidates worth validating further within the Hybrid/OL+BDC, Architecture B cell where this manuscript’s other recommendations already concentrate, not general-purpose replacements for Exposure-weighted or Unadjusted.

4.6 Scope, related work, and limits

This manuscript deliberately narrows the full-scope grid to the covariance-moderation architecture and to the non-linear decay forms, in the interest of a self-contained evaluation of manageable length. The archived full-scope drafts (`archive/report.Rmd` and `archive/report-slim.Rmd`) show that the exposure-weighted advantage documented here does not carry over to mean moderation, where the biomarker moderates the mean response directly and Exposure-weighted is marginally dominated in every design examined. Readers who need that cross-architecture comparison should consult those drafts rather than extrapolate from this one. Establishing what the originally published power values (Section 3.3) were computed under remains outstanding work.

The crossover tradition surveyed by Jones and Kenward[4] would lead one to expect a shape-free lagged term to outperform a parametric predictor once the decay shape is mis-specified. Our results do not support that expectation over the range examined, for two reasons. The exposure-weighted penalty for an incorrect half-life is too small to be overtaken (Section 3.7), and more fundamentally the lagged term does not improve on ignoring carryover at all (Section 4.1), leaving no advantage for decay-shape mis-specification to erode.

One adjacent specification bounds the negative result of Section 4.1. Every carryover term considered there, lagged and washout-adjusted alike, enters as a nuisance main effect with no product against the biomarker. That finding is therefore that a carryover *main effect* does not recover the attenuated signal, not that no shape-free specification can. The natural candidate is a carryover term interacted with the biomarker, either `Sx ~ bm + t + Db + bm:Db + L + bm:L` or its washout-adjusted counterpart with `bm:tsd`, letting the biomarker effect decline across the washout without committing to a rate.

An earlier version of Block S7 tested exactly these two specifications and found a clean negative result at every cell examined. We traced this to a structural cause: this manuscript’s DGP has a single true decay parameter, so a second, interacted carryover coefficient adds estimation noise without a corresponding signal to recover. They were replaced by two specifications addressing the same open problem, the analyst’s unknown decay half-life, from different angles: `G4` (AIC-selected half-life) and `G5` (a paired-difference regression with no carryover representation at all).

Both beat Unadjusted at the Hybrid, Architecture B reference cell, `G4` even edging out Exposure-weighted itself (Block S7). Neither survives a change of trial design or architecture (Blocks S8-S9). `G4`’s AIC selection fails specifically under sparse off-drug sampling (Section 4.5), and `G5`’s advantage appears to depend on the repeated-measures structure it discards being otherwise underused, which is design-dependent. Block S8 shows the Lag-adjusted null of Section 3.4 replicates under Architecture A, so it is not an artifact specific to covariance moderation. Exposure-weighted's advantage, and now `G4`'s, are architecture-sensitive, reversing in direction, though not substantially, under Architecture A (Section 3.7, Block S8).

Three further N-of-1 analysis approaches sit outside the G1-G9 family and are not evaluated here, because each requires substantial independent methodological development rather than a drop-in specification within our existing pipeline.

Tang and Landes[10] propose formula-based t-tests that correct a single-subject paired-difference test's standard error and degrees of freedom for an estimated first-order serial correlation, the natural refinement `G5` lacks (`G5` discards within-patient correlation entirely). Their tests have no biomarker-moderation term, though, and would need a `diff ~ bm` extension with a serial-correlation-aware variance to enter our comparison.

Schmid and Yang[11] describe a Bayesian hierarchical alternative to the frequentist mixed-model and sandwich-estimator family spanning `G1-G9`, with partial pooling of individual-level treatment-effect, trend, and autocorrelation parameters across patients. They note that carryover is tractable only in aggregated, multi-individual analyses like ours, but a biomarker-moderation extension of their model is a separate undertaking.

Daza[12] frames N-of-1 inference as a counterfactual, time-varying-treatment problem, estimated via g-computation or inverse-probability weighting. This is a structurally different paradigm from `G1-G9`'s explicit exponential-decay or lag covariates: carryover there is confounding to be controlled rather than a predictor to be modeled.

None of the three is a candidate for a tenth specification within this manuscript's scope, but each identifies a direction future work on this estimand should consider.

Two further limitations bound what we can conclude. The numbers presented here come from a single parameter set calibrated to a prazosin and PTSD trajectory, so absolute power values should be read as illustrative and the ranking, itself conditional on trial design, as the more portable result. And the sensitivity checks vary one factor at a time and can therefore miss genuine interactions between factors, for instance between high autocorrelation and heavy dropout. The one joint check available to us, block S5 in the full-scope manuscript, revealed no such interaction.

5 Conclusions

1. **The lagged-treatment term does nothing.** It never improved on ignoring carryover altogether, on any measure, in any design, at any half-life, and in every sensitivity block, a clean null given the two specifications share an estimand and differ by a single term. The term enters as a nuisance main effect only, leaving the attenuation of the interaction untouched, and should not be reported as an adjustment for carryover.
2. **Exposure weighting helps, but conditionally.** It achieves the highest power, about 10 points over the unadjusted benchmark, only in the Hybrid and OL+BDC designs, and its lead survives mis-specified decay shape and half-life there. Under crossover it is markedly worse than the binary indicator (0.488 against 0.830) and should not be used.

Where used, we recommend a CR2 cluster-robust standard error (Block S6).

3. **An approximate half-life is adequate, but err short.** A fourfold error at a true half-life of one week costs 6.6 points of power, so a rough pharmacokinetic estimate suffices in practice. The error is asymmetric. Under-assuming drives the predictor toward the still-serviceable binary indicator, while over-assuming erodes the advantage to a tie. Outside block S2 the analyst is granted the true half-life, so the Tier 1 margins are upper bounds.

4. **Dropout matters more than the choice of method.** Power falls sharply as dropout increases, roughly halving at 20%, equally across all three specifications. Preventing dropout is a more effective use of design effort than any choice among them.

5. **The interaction test is somewhat conservative, and correctably so.** All three specifications reject slightly less often than nominal under the null. A CR2 cluster-robust standard error restores the correct rate while recovering modest power, and the exposure-weighted advantage is held or widened under CR2 at four of five S6 stress cells (Block S6).

6. **This is a narrowed slice of a larger study.** Mean moderation and linear decay fall outside this manuscript's scope. The archived drafts `archive/report.Rmd` and `archive/report-slim.Rmd` are the authority on those comparisons and should be consulted before generalizing beyond covariance moderation. We also make no claim of numerical agreement with the originally published power values (Section 3.3).

7. **Carryover terms interacted with the biomarker do not help; two alternative-paradigm specifications do, but only at the reference cell.** Crossing a lagged or washout carryover term with the biomarker gave lower power than the unadjusted benchmark at every cell examined (Block S7's original version). We traced this to a structural cause: this manuscript's DGP has a single true decay parameter, so a second, interacted coefficient adds estimation noise without a corresponding signal to recover (item 1's mechanism, restated). They were replaced by two specifications answering the same open question, the analyst's unknown decay half-life, differently. `G4` estimates it by AIC selection; `G5` discards the decay representation, and the mixed model itself, for a two-stage paired-difference regression following Senn, Julious and Araujo[7]. Both beat Exposure-weighted at the Hybrid, Architecture B reference cell (Blocks S7-S8; `G4` 0.850, `G5` 0.842, `G3` 0.836 at $t_{1/2} = 1.0$). Neither result generalizes: Block S8 shows both reverse, like Exposure-weighted itself, under Architecture A, and Block S9 shows both fail under the crossover design, `G5` mildly and `G4` catastrophically. `G4`'s CO failure has a clear mechanism: AIC selection systematically drifts to the boundary of its candidate half-life grid when off-drug sampling is sparse and widely spaced, reproducing the S2 finding that over-assuming the half-life is the more damaging error, now through the estimation procedure itself rather than a fixed wrong assumption. Both specifications should be read as promising candidates for further validation at the one cell type where this manuscript already recommends Exposure-weighted, not as general-purpose replacements for it or for Unadjusted.

8. **Adding CR2 to the AIC-selected specification gives the single best result in this manuscript, at one cell only.** `G9` (`G4` plus a CR2 cluster-robust standard error) outperforms all eight other specifications at the Hybrid, Architecture B reference cell, including `G3`+CR2, and its Type I error holds at nominal despite the post-selection risk `G4`'s half-life selection step raises in principle (Section 3.7, Blocks S10-S11). We read this as a promising direction, not a recommendation: `G9` has not been tested against decay-shape mis-specification, dropout, a crossover design, or an alternative architecture, all of which item 2's specification (`G3`+CR2) has already been checked against.

6 Conflict of interest

The authors declare no conflicts of interest.

7 Funding

This work received no specific funding.

8 Data availability statement

The data and code supporting the findings of this study are available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`. Simulation drivers are at `analysis/scripts/carryover-sensitivity/`. Exact paths for this manuscript are listed in the Reproducibility section below.

9 Reproducibility

This manuscript was rendered with R 4.6.0 inside the project's zzcollab Docker container. The production simulation ran under R 4.5.3 (§3.6), with the image hash in `Dockerfile` and the package set in `renv.lock`. The simulation study uses `nlme` for the continuous-time linear-mixed analysis, `parallel` for the 8-core parameter-grid sweep, and `data.table` or `furrr`/`purrr` in the two simulation collections. No simulation code runs during knitting, which loads pre-computed checkpoints only. Each driver sets a master seed (20260507) and derives per-replicate seeds from it via `parallel::mc.set.seed`.

Data and code availability. All simulation outputs, cell-level summaries, and driver scripts are versioned under `analysis/scripts/carryover-sensitivity/` and `analysis/data/` in the pmsimstats-ng repository. This includes the principal factorial, the S1 to S5 sensitivity blocks, the S6 cluster-robust recalibration, the washout-adjusted (E4) check of Section 4.1, and the high-precision prototype rerun of Section 3.7. It also includes the S7 alternative-paradigm block (`09-run-sensitivity-s7.R`, `output/09-sensitivity-s7.rds`); the S8 architecture-by-half-life block (`10-run-sensitivity-s8.R`, `output/10-sensitivity-s8.rds`); the S9 CO-design block (`14-run-sensitivity-s9.R`, `output/14-sensitivity-s9.rds`); the S10

CR2 comparison block (`15-run-sensitivity-s10.R`, `output/15-sensitivity-s10-g.rds`); and the S11 Type I error block (`16-run-sensitivity-s11.R`, `output/16-sensitivity-s11.rds`). All five were added 2026-08-18/19 (master seed 20260415, 500 replicates per cell except S11's 2000; Hybrid reference cell except S9, which uses CO, and S11, which sets $c_{bm} = 0$). `E5`/`E6` (biomarker-interacted carryover terms) were tested in an earlier version of S7-S9 and are documented in Section 4.6; they were replaced by `G4` (AIC-selected half-life, reusing the candidate-grid logic of `characterize_carryover()`, `R/carryover_analysis.R`) and `G5` (paired-difference regression). The `E1`/`E3`/`E2`, `E7`, `E9` fitters, and the `E1cr2`/`E3cr2`/`E7cr2` CR2 variants used by S10 (`cr2_extract()`, `fit_spec_s7`, `simulate_cell_s7`), are appended to `simulation-core.R` alongside the Tier 1 `fit_spec`/`fit_all_specs` pair, which they do not modify. `spec-labels.R` additionally defines the `G1`-`G9` compact display-code layer (`g_code`, `g_labels`) used in S10's table and figure; `G8` (`G3`+CR2, stored as `E2cr2`) is reused directly from Block S6 rather than rerun, so it does not share common random numbers with the rest of S10's comparison. Figure 4, and the decay-form mismatch panel (Figure 6), retain the original three specifications (G1-G3), since decay shape is the only axis those two figures vary. Both were regenerated from a dedicated 18-cell simulation at a more extreme Weibull shape range ($k = 0.5, 2.0$, replacing an earlier $k = 0.7, 1.5$; $k = 1.0$ is not a separate condition, since it duplicates Exponential exactly) than the shared production grid (`02-grid-summary.rds`) still uses (`23-run-decay-shape-sensitivity.R`, `02-decay-shape-sensitivity.rds`, $n_{sim} = 500$, Architecture B only). This is a standalone script rather than a change to `01-run-factorial.R`'s shared grid, since that grid is also consumed by the archived full-scope drafts and other manuscripts. Figures 2 and 3 were later expanded to all nine G1-G9 specifications, from a dedicated 30-cell simulation restricted to the exponential DGP and Architecture B (`19-run-tier1-hendrickson-g9.R`, `02-grid-summary-hendrickson-g9.rds`, preliminary $n_{sim} = 100$). This replaced the original three-specification version of both panels rather than supplementing it, and superseded the separate Type I error checks the manuscript previously reported at Figures 6-9 (the Tier 1 Type I boxplot/heatmap and the S6 G1-G3/G1-G9 Type I dot plot/heatmaps), since the $c_{bm} = 0$ column of Panel A now reports Type I error for all nine specifications directly. Those retired figures' underlying data and scripts (`03-render-figures-extra-slim.R`, `07-type1-cr2-figure-extra-slim.R`, `18-run-sensitivity-s6-g9.R`) are retained for archival purposes but no longer referenced by this document. The corresponding Tier 2 heatmaps (S1-S4, S6, S7, S8, S9, S10, S11) are produced by `13-hendrickson-heatmap-tier2.R` from `02-sensitivity-summary.rds` and the S7/S8/S9/S10/S11 output `.rds` files (plus `diag-s6-cr2.rds` for `G8` and Table 4); these Tier 2 figures are additional to, not replacements for, the existing S1-S4 line plots and the S7/S8 kable tables. The grid design and Morris ADEMP pre-registration are in `simulation-grid-plan.md` in the same directory. Superseded wider-scope drafts are retained under `archive/`, documented in its `README.md`.

Specification labels. All nine specifications are stored under `E`-prefixed codes (`E1`, `E2`, `E3`, `E7`, `E9`, `E1cr2`, `E2cr2`, `E3cr2`, `E7cr2`) in every data file and script, and this manuscript displays them uniformly as `G1`-`G9`, or by name, at render time only. The mapping is applied by `spec-labels.R`, so the stored values and archived outputs are unaffected: what a reader sees as `G3` (Exposure-weighted) in the prose above is stored as `E2` in every `.rds` file this

1073 manuscript reads. Output archived before 2026-07-31 stores the specifications as `A1`, `A2`, `A3`.
1074 The migration is documented in `analysis/report/NOTATION.md`.

1075 10 References

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